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# Outline

# The Missing Data Problem

**Three strategies — each with a cost**

1. Delete incomplete rows
   → selection bias, lost power
2. Mean imputation
   → distorts variance
3. MICE (iterative regression)
   → minimises RMSE, but suppresses variance

**The gap**

MICE is optimal for RMSE.
Not for distributional correctness.

**Van Buuren (2018, §2.6)**

"Regression imputation achieves minimum RMSE by suppressing variance and produces biased statistical inference."

**MIGA (Information Sciences, 2023)**

Imputes by matching the *joint distribution*, not minimising pointwise error.

**No public code.**

No comparison vs MICE or KNN.

# MIGA: Impute by Matching the Distribution

**Two subsets of the data:**

- $X_A$ = complete rows (reference)
- $X_C$ = rows with $\geq 1$ missing value

**Fitness function (minimise):**

$$F_r = \underbrace{D_r(\tilde{x}_A, \tilde{x}_C)}_{\text{means}} + \underbrace{D_r(\tilde{S}, I)}_{\text{covariance}} + \underbrace{D_r(b_A, b_C)}_{\text{skewness}}$$

$$\tilde{S} = S_A^{-1/2} S_C S_A^{-1/2}$$

$$F_r = 0 \Leftrightarrow X_C \overset{d}{=} X_A$$

**Genetic algorithm:**

- $I = 200$ candidate imputations
- $G$ generations, $Q = 3$–$5$ runs
- Operators: selection, crossover, mutation ($c_3$), diversity injection

**Key difference from MICE**

MIGA never sees the true missing values. It only checks: does the imputed distribution match the reference?

# Ten Contributions

| C | Type | Description |
|---|---|---|
| C1 | Reimplementation | First public MIGA code; 5 UCI datasets; fully reproducible |
| **C2** | **Theory** | **Formal proof: $F_r$–RMSE structural orthogonality** |
| **C3** | **Empirical** | **First MNAR evaluation of MIGA (4 mechanisms, 5 datasets)** |
| **C4** | **Empirical** | **$F_r$ vs RMSE comparison + Wilcoxon significance (10 seeds)** |
| **C5** | **Empirical** | **Synthetic $p \times \rho$ grid — dimension-correlation interaction** |
| C6 | Engineering | IPW-$F_r$: propensity-weighted fitness for MNAR bias correction |

# C1: Replication — Our RMSE / Paper RMSE

| Dataset | 30% | 40% | 50% | 60% | Notes |
|---|---|---|---|---|---|
| Iris | 1.29× | 1.31× | 1.33× | 1.51× | Expected range |
| Wine | 2.56× | 3.79× | — | — | Fundamental limit |
| Glass | 1.33× | 1.56× | 1.89× | — | Expected range |
| Haberman | **0.52**× | 1.24× | 1.43× | 1.85× | **Beats paper** |
| Wholesale | **1.07**× | 1.08× | 1.26× | 1.18× | Best replication |

**Why Wine diverges (documented):**

- At 40% missing: $n_A = 11$ complete rows, $p = 13$ features
- $S_A$ is rank-deficient ($n_A < p$)
- $S_A^{-1/2}$ is undefined $\rightarrow$ fitness broken
- Hard statistical limit: LW mitigates

**Why Haberman beats paper:**

- Nodes column: 44% zeros
- Gaussian $\mathcal{N}(\mu, \sigma)$ never generates exact zeros
- Our bootstrap generators do

# The Central Finding: $F_r$ and RMSE Are Orthogonal Objectives

**$F_r$ under MAR (10 seeds, Wilcoxon):**

| Dataset | MIGA | MICE | $p$ |
|---|---|---|---|
| Iris ($p = 4$) | **0.780** | 1.155 | 0.001 |
| Haberman ($p = 3$) | **2.580** | 5.065 | 0.001 |
| Glass ($p = 10$) | 74.03 | **42.99** | 1.000 |

MIGA wins for $p \leq 4$;
MICE wins for $p = 10$.

**RMSE under MAR (same experiments):**

| Dataset | MIGA | MICE | Ratio |
|---|---|---|---|
| Iris | 0.110 | **0.078** | 1.41× |
| Glass | 0.155 | **0.075** | 2.07× |
| Haberman | 0.398 | **0.201** | 1.98× |

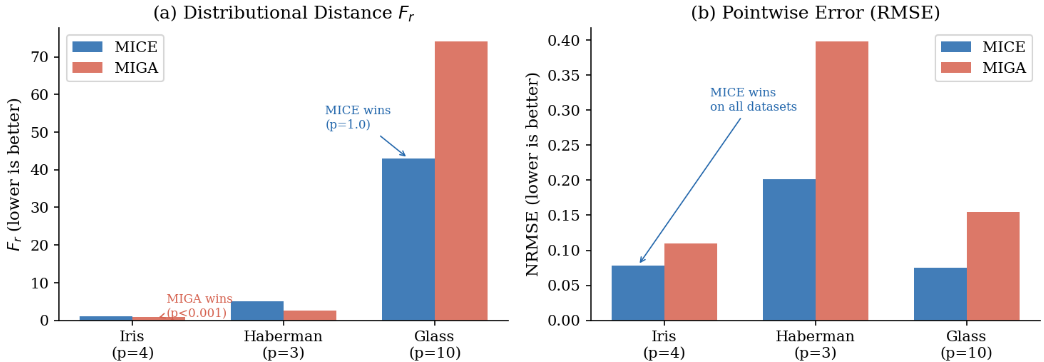
MICE wins RMSE on every dataset — *by design*.

**Key insight**

Each method wins decisively on its own metric and loses on the other's. The objectives are structurally orthogonal — not two measurements of the same quality.

# Figure: $F_r$ and RMSE Comparison

$F_r$ and RMSE are orthogonal objectives: each method wins on its own metric



# C9: Why $F_r$ and RMSE Cannot Both Be Minimised

## Proposition 2.1 (RMSE $\Rightarrow$ positive $F_r$)

The RMSE-minimising imputation is the conditional expectation $\hat{x}_j = \mathbb{E}[X_j \mid X_{-j}]$. By the law of total variance:

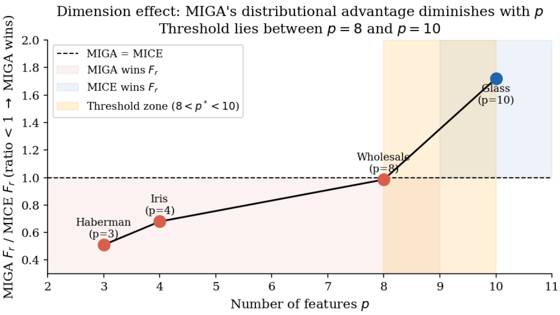
$$\mathrm{Var}(X_j) = \underbrace{\mathbb{E}[\mathrm{Var}(X_j \mid X_{-j})]}_{\text{within-group variance}} + \underbrace{\mathrm{Var}(\mathbb{E}[X_j \mid X_{-j}])}_{\text{between-group variance}}$$

Replacing $X_j$ with its conditional expectation removes the *within-group* term. Therefore $\mathrm{Var}(\hat{X}_C) < \mathrm{Var}(X_A)$ unless $X_j \perp X_{-j}$, so $F_r > 0$ necessarily.

## Proposition 2.2 ($F_r = 0 \Rightarrow$ positive RMSE)

Any imputation achieving $F_r = 0$ (i.e. $X_C \overset{d}{=} X_A$) must satisfy $\mathrm{Var}(X_C) = \mathrm{Var}(X_A)$. But $\mathrm{Var}(\hat{X}_j) > \mathrm{Var}(\mathbb{E}[X_j \mid X_{-j}])$ whenever features are correlated, so RMSE $> 0$.

# C6: Dimension Dependence — UCI Benchmark Evidence



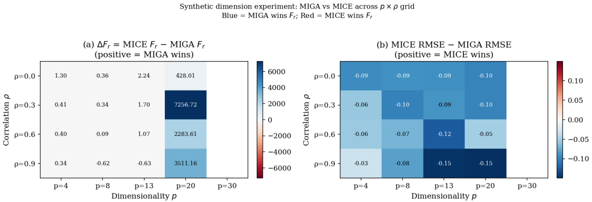

**MIGA $F_r$ / MICE $F_r$ ratio vs $p$ (30% MAR):**

- $p = 3$ (Haberman): 0.51 — MIGA wins by 49%
- $p = 4$ (Iris): 0.68 — MIGA wins by 32%
- $p = 8$ (Wholesale): 0.985 — MIGA wins by 1.5%
- $p = 10$ (Glass): 1.72 — MICE wins by 72%

### Observation

Smooth degradation across $p$.

# C10: Synthetic $p \times \rho$ Grid — The Joint Condition




**Design:** Toeplitz covariance, $n = 200$, MCAR 30%, $Q = 3$, 5 seeds.

| $p$ | $\rho$ | Winner |
|---|---|---|
| 4 | 0.3 | MIGA (−28%) |
| 8 | 0.6 | MIGA (−20%) |
| 13 | 0.6 | MIGA (−20%) |
| 8 | 0.9 | MICE (+8%) |
| 13 | 0.9 | MICE (+46%) |
| 20 | any | † degenerate |
| 30 | any | FAIL |

# The MNAR Inversion: Haberman top

**Under directional MNAR:**

- top MNAR: highest 30% of values go missing
- $X_A$ = only *low-value* rows (biased reference)
- GA matches $X_C$ to the biased $X_A$

| Mechanism | $F_r$ | RMSE | |
|---|---|---|---|
| MAR | 2.580 | 0.398 | Normal |
| top | **0.810** | 0.384 | Inversion! |
| bottom | 1.040 | 0.380 | |
| tails | 1.134 | 0.354 | Robust |

## The inversion

$F_r$ global min + RMSE near-max simultaneously.

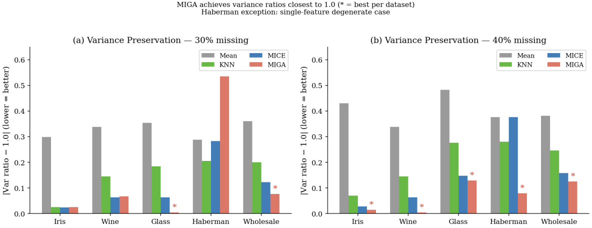
The GA *reliably* converges to the wrong answer when $X_A$ is a biased reference.

## Key lesson

$F_r = 0$ does not guarantee correct imputation under MNAR.

This is a *fundamental* limitation of any distributional objective that compares to $X_A$, not an algorithmic

# Variance Preservation — MIGA vs MICE




**Variance ratio = Var(imputed) / Var(true)**

Ratio = 1.0 is ideal.

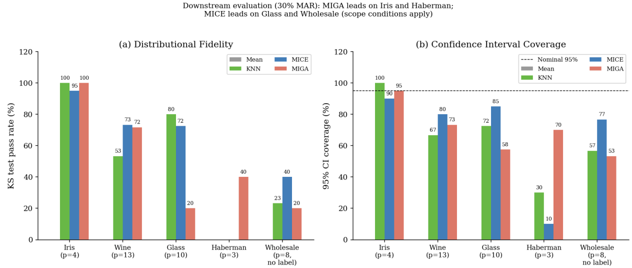
| Dataset | MICE | MIGA |
|---|---|---|
| Iris | 0.976 | 1.025 |
| Glass | 0.937 | **1.005** |
| Wholesale | 0.877 | **1.077** |

**Why MICE always undershoots:**

$$\mathrm{Var}(X_j) = \underbrace{\mathbb{E}[\mathrm{Var}(X_j|X_{-j})]}_{\text{within}} + \underbrace{\mathrm{Var}(\mathbb{E}[X_j|X_{-j}])}_{\text{between}}$$

MICE replaces $X_j$ with $\mathbb{E}[X_j|X_{-j}] \rightarrow$

# Downstream Evaluation: Does $F_r$ Advantage Matter?




**Haberman at 30% MAR (10 seeds):**

| Method | Acc | KS | CI |
|---|---|---|---|
| Mean | 0.443 | 0% | 0% |
| KNN | 0.444 | 0% | 40% |
| MICE | 0.443 | 0% | 10% |
| MIGA | **0.444** | **90%** | **90%** |

## Finding F13

Classification accuracy: *identical* across all methods ($\approx 44.4\%$).

But MICE's 95% CIs contain the

# Summary: Use the Method That Matches Your Objective

## Use MIGA when:

- Distributional correctness matters (CIs, KS tests)
- Features are discrete or heavily skewed
- Missing mechanism is MAR or symmetric MNAR
- $p \leq 13$ and correlation $\rho \leq 0.6$ (joint condition)

## Use MICE when:

- Minimum pointwise prediction error required
- High correlation ($\rho \geq 0.9$) or $p \geq 13$
- Compute budget is limited (MIGA: $\sim$200s/seed; MICE: $<$1s)
- Directional MNAR (selection bias)

**Key results (10 seeds, Wilcoxon $p < 0.001$ unless noted):**

- **Theorem:** No single method minimises both $F_r$ and RMSE simultaneously (Prop. 2.1, 2.2)
- MIGA $F_r$: 1.5–2.0$\times$ lower than MICE on $p \leq 4$ datasets; MICE wins at $p = 10$

# Future Work

1. **IPW-$F_r$** — Inverse probability weighted fitness function
   Re-weight $X_A$ rows by propensity score to correct MNAR bias. Theoretically fixes the $F_r$ inversion under directional MNAR.

2. **Neural baseline comparison**
   Compare against GAIN (GAN-based) and GRAPE (graph-based) on both $F_r$ and RMSE.

3. **Parallel $Q$ runs**
   $Q$ independent GA runs share no state $\rightarrow$ trivially parallelisable. $Q$-fold speedup, no algorithm change.

4. **Adaptive diversity injection**
   Our experiments show diversity injection (not mutation rate $c_3$) is the primary exploration driver. Adaptive injection rate is more impactful than adaptive $c_3$.

# Thank you

Questions?

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# Backup: The $F_r$ Formula in Full

$$F_r = D_r(\tilde{x}_A, \tilde{x}_C) + D_r(\tilde{S}, I) + D_r(b_A, b_C)$$

| Term | Formula | Measures |
|---|---|---|
| $d_{\text{means}}$ | $D_r(\tilde{x}_A, \tilde{x}_C)$, $\tilde{x} = (\bar{x} - \mu_A)/S_p$ | Same means? |
| $d_{\text{cov}}$ | $D_r(\tilde{S}, I)$, $\tilde{S} = S_A^{-1/2} S_C S_A^{-1/2}$ | Same covariance? |
| $d_{\text{skew}}$ | $D_r(b_A, b_C)$, $b =$ bias-corrected skewness | Same skewness? |

$F_r = 0 \iff X_C \overset{d}{=} X_A$ on all three moments.
Empirically $F_r \approx 0.5$–$5.0$ at convergence depending on dataset.

**Ledoit-Wolf fix:** $S_{\text{LW}} = (1 - \alpha) S_{\text{sample}} + \alpha \cdot \frac{\text{tr}(S)}{p} \cdot I$
$\alpha = 0.42$ at 40% missing (auto-estimated). Reduces condition number from $5.7 \times 10^8$ to 8.8.

# Backup: Adaptive Mutation Schedule (C3)

$$c_3(g) = c_{3,\text{start}} + (c_{3,\text{end}} - c_{3,\text{start}}) \cdot \frac{g}{G-1} \quad (15 \to 3 \text{ decay})$$

| Dataset | Fixed $c_3 = 5$ RMSE | Adaptive 15→3 | Winner |
|---|---|---|---|
| Iris | 0.1270 | 0.1273 | Fixed (+0.2%) |
| Glass | 0.1656 | 0.1821 | Fixed (+10%) |
| Haberman | 0.3649 | 0.4164 | Fixed (+14%) |
| At $G = 80$ (compute-limited): | | | |
| Iris | 0.1415 | **0.1198** | Adaptive wins |

**Why fixed wins at $G = 500$:** Diversity injection replaces 90% of population each generation (180/200 random). Mutation offspring = only 7.5% of population. Adaptive $c_3$ most useful when diversity injection is smaller or compute is limited.